



江苏省苏州中学国际部  
Suzhou High School – International Division

# SZZX-ID BP Physics

## Electromagnetic Induction

# Class Outline

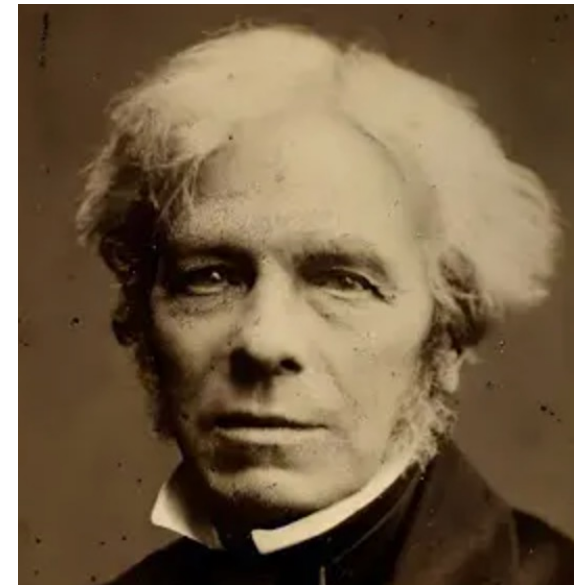
|                            |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Title of Unit</b>       | Magnetic field                                                                                                                                                                                                                                                                                                                               |
| <b>Title of Lesson</b>     | Electromagnetic Induction                                                                                                                                                                                                                                                                                                                    |
| <b>Starter Activity</b>    | Video: induced current demo                                                                                                                                                                                                                                                                                                                  |
| <b>Direct Instructions</b> | Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry                                                                                                                                                                                                                                                             |
|                            | <ul style="list-style-type: none"> <li>- Inquiry: experiment by a simulator</li> <li>- Lecture: Faraday's law and right hand rule</li> </ul>                                                                                                                                                                                                 |
|                            | Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher                                                                                                                                                                                                                                                         |
|                            | - Students are advised to choose the tool (one of the two right-hand rules) that they are comfortable with and practice                                                                                                                                                                                                                      |
|                            | International Mindedness:                                                                                                                                                                                                                                                                                                                    |
| <b>Lesson Objectives</b>   | <p>By the end of the lesson, students will be able to:</p> <ul style="list-style-type: none"> <li>- explain how an electromotive force (e.m.f.) is induced by moving a conductor through a magnetic field</li> <li>- understand the direction of forces, fields and current in electromagnetic induction by using right hand rule</li> </ul> |
| <b>ATLs</b>                | Identify unit ATLs and how they are being addressed in the lesson                                                                                                                                                                                                                                                                            |
|                            | Inquiry                                                                                                                                                                                                                                                                                                                                      |
| <b>IB Learner Profile</b>  | Link the lesson to the characteristics of the IB Learner attribute                                                                                                                                                                                                                                                                           |
|                            | Inquirer, thinker                                                                                                                                                                                                                                                                                                                            |
| <b>Teaching Activities</b> | <ul style="list-style-type: none"> <li>- Video: induced current</li> <li>- Simulation: virtual experiment</li> <li>- Example questions to practice left hand rules</li> </ul>                                                                                                                                                                |
| <b>Plenary Session</b>     | Checking for understanding / Homework / Q & A / Reflection etc.                                                                                                                                                                                                                                                                              |
|                            | Differentiate left and right hand rules (left-hand grip – field around a wire and a solenoid; left-hand rule: force on a current-carrying wire in a field; right-hand rule (today's) – direction of induced current)                                                                                                                         |

# Brief history of electromagnetic induction



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- By the end of the 18th century, scientists had noticed many electrical phenomena and many magnetic phenomena, but most believed that these were distinct phenomena
- In 1831, Michael Faraday discovered electromagnetic induction through his famous induction ring experiment, where a changing magnetic field induced e.m.f., hence a current in a circuit – electromagnetic induction



# Faraday's law of induction



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- Faraday's law: An e.m.f. will be induced in a conductor in the presence of a changing magnetic field.
- Faraday concluded a few ways to induce an emf in a secondary coil, by
  - A changing current
  - A changing magnetic field
  - A constant current in a moving conductor
  - A moving magnet

# Activity – simulate induced current



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- Open the simulator (MB - lesson 7 - stream and resources)

Try the following

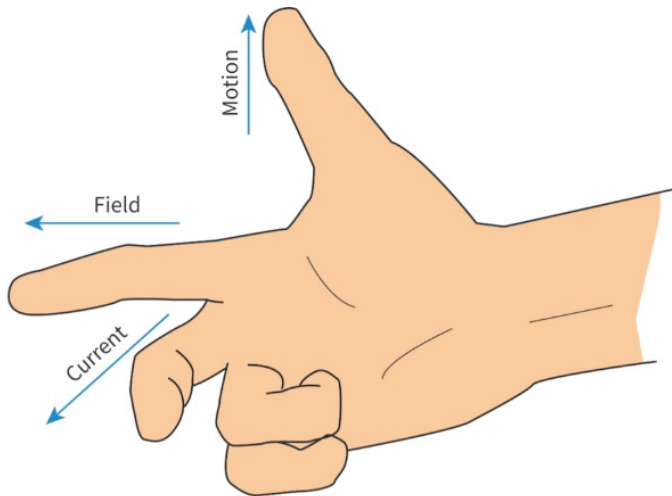
- Move the magnet in and out of the coil at different speeds and observe the brightness of the bulb
- Add a voltmeter, repeat step 1, observe the magnitude and sign of the voltage.
- Try the two coils with different numbers of turns, observe the voltage
- Put the magnet in the coil without moving it, observe the voltage

Two ways to induce an e.m.f.  
1. Moving a magnet near a wire, or  
2. Moving a wire across a magnetic field  
So that so that the magnetic field lines are cut by a wire

# Fleming's right hand rule – determine the direction of induced current



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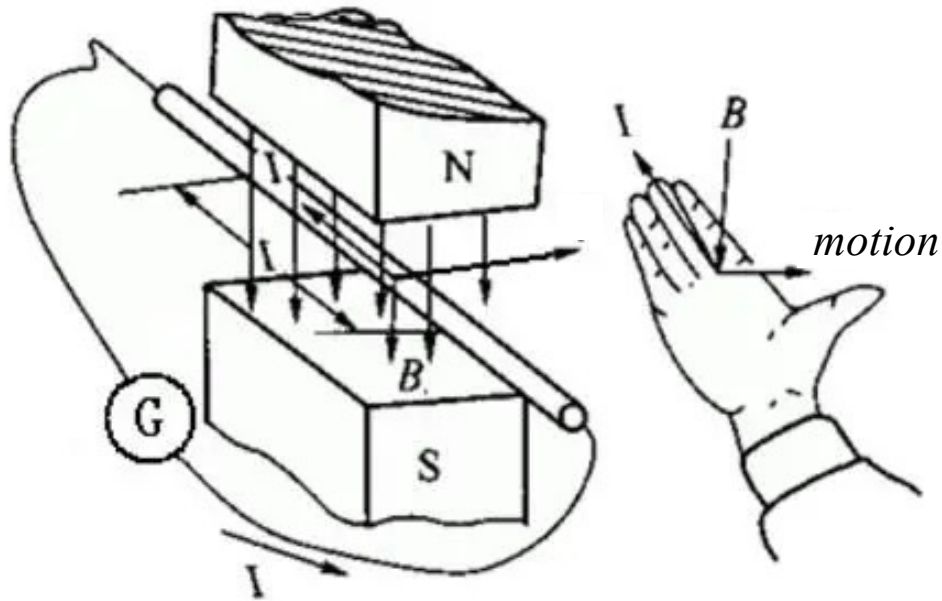
When an e.m.f. is induced in a wire, use the right-hand rule to work out the direction of motion, magnetic field and induced current

- The direction of **M**otion of the wire (relative to the field) is represented by the thu**M**b.
- The direction of the magnetic **F**ield is represented by the **F**irst finger.
- The direction of the **C**urrent is represented by the se**C**ond finger.

# Another version of right-hand rule to determine induced current



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use your right hand

- Thumb pointing to the direction of motion
- Let B field lines go through your palm
- The four fingers points to the direction of the current

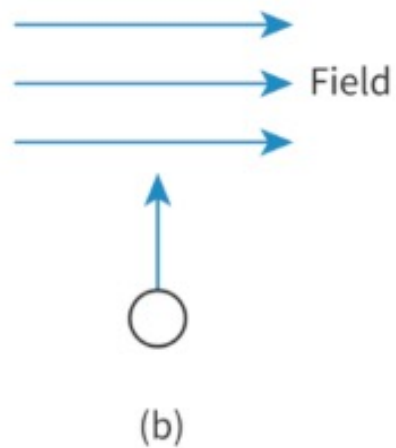
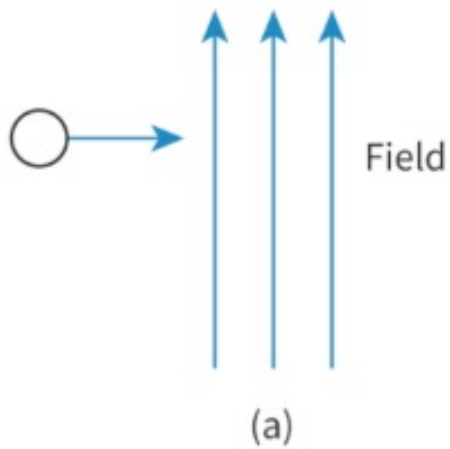
Pick one of the ways to determine current, not both!

# Example



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The circles represent the cross-section of a wire. The wire is moving in the direction shown by the single arrows. For (a) and (b), determine the direction of the induced current (either into or out of the page).



Follow the guidance about using the right-hand rule to see if you agree with the following answers.

(a) Field: up. Motion: right

Current: out of page

(b) Field: right. Motion: up

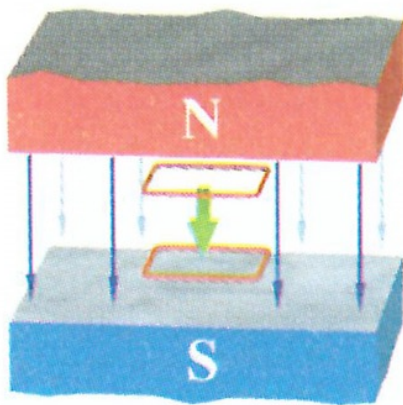
Current: into page

# Exercise

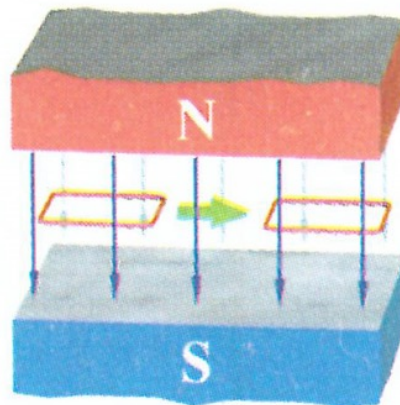


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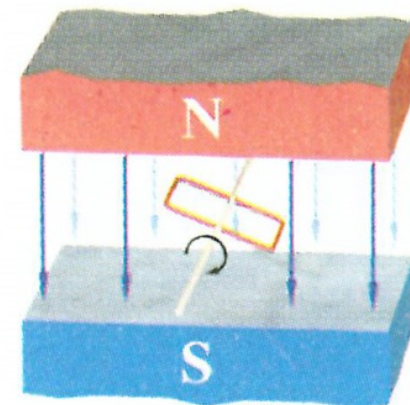
A closed circuit moves in a uniform magnetic field. In which of the three circumstances below will current be induced in the circuit?



A



B



C

# Exercise



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An expandable circuit is stretched and placed in a uniform magnetic field. Once it's released, it will contract. Will there be current induced in the circuit? Explain why?

